UNIVERSITI SAINS MALAYSIA

SCHOOL OF COMPUTER SCIENCES

CDS504 – Enabling Technologies & Infrastructures for Big Data
Semester 2, Session 2019/2020

Assignment 1

Objective To simulate Hadoop MapReduce usage on some open/free dataset to obtain interesting results.

Tasks

i) Identify a problem/job which can be solved using MapReduce.
ii) Find a suitable open/free dataset. *(E.g. kaggle.com, data.world, data.gov, data.gov.my, dosm.gov.my)*
iii) Extract and reformat the data you're interested in, if needed.
iv) Show how the various stages of MapReduce (i.e. splitter, mapper, shuffler/sorter, reducer) will process your data. The reducer should implement at least 2 different functions (i.e. total, average, median, maximum, minimum, standard deviation, etc.).
v) Show how new/interesting results could be obtained from the dataset (which did not exist originally in the dataset) using MapReduce.

Hint/Suggestion

Use WordCount example as basis, but develop a new solution which is different, and using your own dataset.

Report

Prepare a written report which include the following:

- Problem statement (i.e. what is the issue), motivation (i.e. why is the issue important, why MapReduce is suitable, etc.) and the goal of your work (i.e. what is the expected end-result, how it will be useful, etc.).
- Selection and description of the dataset.
- Data pre-processing and reformatting, if any.
- How the mapper and reducer process the data. Show the various stages, with their intermediate results.
- Description and discussion on the final results obtained.
- Reflection (i.e. lessons learned, experience developing solution for MapReduce, etc.).
- References
- Appendix – Dataset (original and extracted/reformatted)

Online Submission

Upload the report (in PDF format) to eLearn@USM. Only one submission per group is required.

Viva

No presentation required.

Important Note

Plagiarism is not tolerated. Direct "cut-and-paste" from Internet or other sources is not acceptable. You must show this is your own work (written using your own words).

All references used must be properly acknowledged. Use the standard format for listing references.

Group: 1-4 persons in a group. Everyone in the group must contribute.

Contribution: 20% of coursework.

Due date: Friday, 8th May 2020, 5.00pm.